

Question (from image_ef60dc.png):

- What is Virtual Memory? Explain how it works, its key mechanisms (such as Page Table), and its primary benefits. / Virtual Memory → Use + Workings

Answer:

- **What it is:** Virtual memory is a technique that uses the hard disk as extra, slower RAM, pretending the system has more memory than physically available. [PDF](#)
- **Primary Benefits (Use):** It allows programs to run even if they require more memory than the actual physical RAM size, letting physical RAM act like a cache for the disk. [PDF](#)
- **How it Works & Key Mechanisms:**
 1. **Pages:** Memory is divided into fixed-size chunks called pages. [PDF](#)
 2. **The Page Table:** It acts like memory's GPS or address book. It stores the mapping that translates the virtual addresses used by programs into real physical addresses in RAM. [PDF + 1](#)
 3. **Translation Process:** When the CPU requests data via a virtual address, the system checks the Page Table. If the page is in RAM (Page Hit), it translates the address instantly. If it is on the disk (Page Fault), the OS pauses the program, loads the page into RAM, updates the table, and resumes execution. [PDF + 2](#)

Question:

- Pages, Pagetable

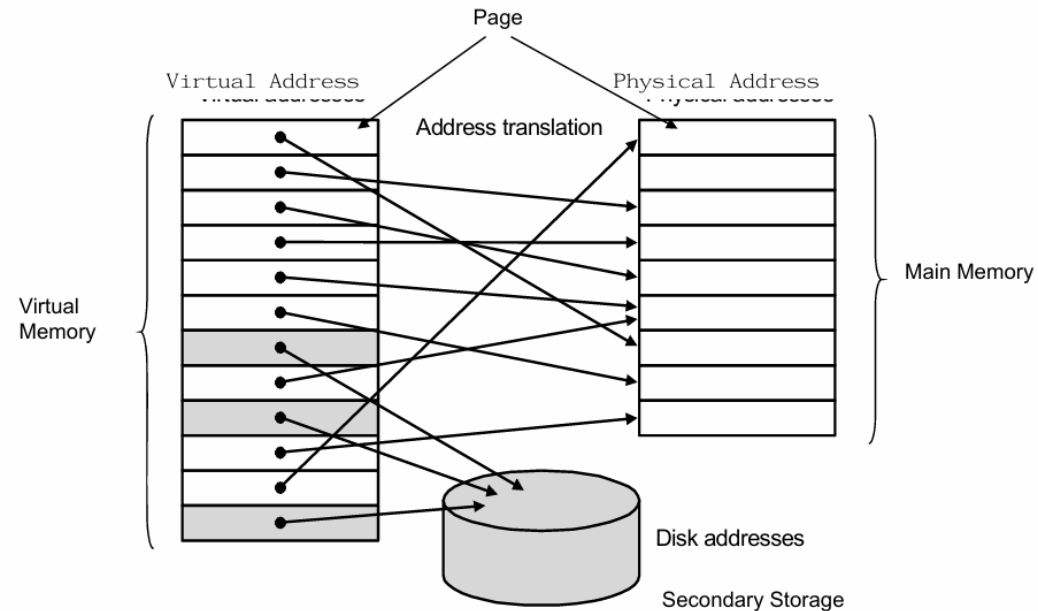
Answer:

- **Pages:** Memory is split into fixed-size chunks called pages (typically 4 KB). Disk-to-RAM data transfers always happen in whole pages. [PDF + 1](#)
- **Page Table:** It acts as memory's GPS or address book. It maps virtual addresses (what programs use) to physical addresses (real RAM locations) so the CPU can find the data.

[PDF + 1](#)

Virtual Memory Diagram

Virtual Memory: A visual



Mapping of pages from a virtual address to a physical address or disk address